Solve the Sum of Subset Problem-Backtracking Method

W [1:4] = {2, 3, 5, 7}, M = 10

STATE-SPACE TREE

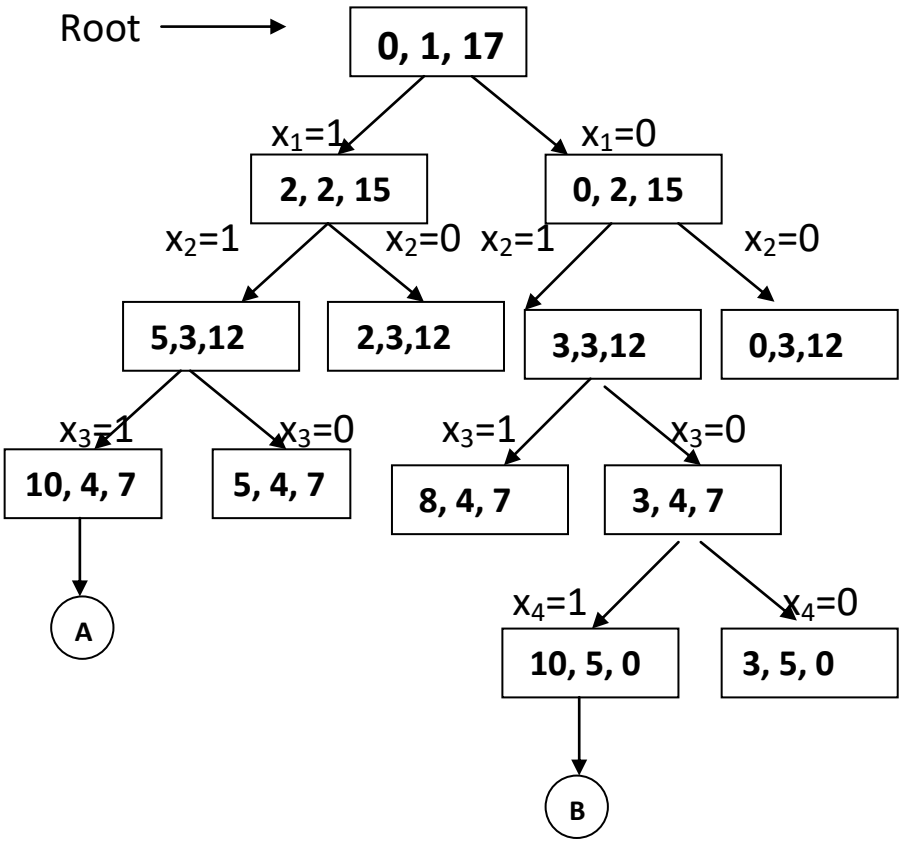
s=0, k=1, r=17

Initial Call - SumofSub (0, 1, 17) - Root

SumOfSub(s + w[k], k+1, r-w[k]); - Left Child

SumOfSub(s, k+1, r-w[k]); - Right Child

W [1:4] = {2, 3, 5, 7}, m=10



Solutions

A ⟶ {1, 1, 1, 0}

$x_1=1, x_2=1, x_3=1, x_4=0$

B ⟶ {0, 1, 0, 1}

$x_1=0, x_2=1, x_3=0, x_4=1$